



ACUTE CARE SURGERY MU HEALTH ANTIBIOTIC PRACTICE MANAGEMENT GUIDELINE

Acute Care Surgery at the University of Missouri is committed to delivering safe, evidence-based, and cost-conscious care across the continuum of trauma, emergency general surgery, critical care, and elective surgical practice. As antimicrobial resistance continues to rise and practice variation persists nationally, establishing a unified antibiotic stewardship framework is essential to optimizing patient outcomes and preserving the effectiveness of critical therapies.

This Practice Management Guideline provides standardized, diagnosis-specific antibiotic recommendations derived from contemporary national guidance including the Infectious Disease Society of America, Surgical Infection Society, the American Association for the Surgery of Trauma, and other authoritative bodies—and aligned with our institution’s local microbiologic trends and antibiogram. By creating consistent expectations for antibiotic selection, timing, dosing, and duration, this PMG aims to reduce unnecessary variation, promote judicious antimicrobial use, and ensure that every patient treated within our trauma and surgical care system receives the safest and most appropriate therapy.

Acute Care Surgery MUHEALTH Antibiotic Practice Management Guideline						
Trauma						
Injury	Antibiotic	Dose	Frequency	Duration	Redose	PNC Allergy/Alternative
Abdominal Trauma with/or without Hollow Viscus Injury	Cefazolin <i>plus</i> Metronidazole	2 g 500 mg	Q8H Q12H	24 hours OR 4 days after source control (consider monotherapy Zosyn)	Every 4 hours or with >1500ml blood loss or > 10 units blood transfusion	Levofloxacin and Metronidazole
Chest Tubes	N/A	N/A	N/A	N/A	N/A	N/A
Craniotomy	Cefazolin	2 g	Q8H	24 hours	N/A	N/A
Open Skull Fracture	Ceftriaxone	2 g	Q12H	24 hours after closure or 72 hours total, whichever is shortest	N/A	Contamination: ceftriaxone and Metronidazole x 72 hours
Penetrating Brain/CNS Injury	Ceftriaxone	2 g	Q12H	24 hours after closure or 72 hours total, whichever is shortest	N/A	Contamination: ceftriaxone and Metronidazole x 72 hours
Ventriculostomy or ICP Monitor	N/A	N/A	N/A	N/A	N/A	N/A
Pneumocephaly	N/A	N/A	N/A	N/A	N/A	N/A
CSF Leak	N/A	N/A	N/A	N/A	N/A	N/A
Open Face Fracture/Mandible Fracture	Ampicillin-sulbactam	3 g	Q6H	24 hours after closure or 72 hours total, whichever is shortest	N/A	Clindamycin 600mg q8 hours
Closed or Non-operative Face Fracture	N/A	N/A	N/A	N/A	N/A	N/A
GU Trauma	Cefazolin <i>plus</i> Metronidazole	2 g 500 mg	Q8H Q12H	24 hours	Every 4 hours or with >1500ml blood loss or > 10 units blood transfusion	Levofloxacin and Metronidazole
Vascular	Cefazolin	2 g	Q8H	24 hours	Every 4 hours or with >1500ml blood loss or > 10 units blood transfusion	Vancomycin – Pharmacy to dose



Emergency General Surgery						
Appendicitis	Ceftriaxone <i>plus</i> Metronidazole	2 g 500 mg	Q24H Q12H	Post Appendectomy, non-perforated: 24 hours Perforated with source control: 4 days OR 8 days in Critical Illness Perforated without source control: 8 days	N/A	Levofloxacin <i>plus</i> Metronidazole or Ertapenem 1 g Q24H
Diverticulitis with Source Control	Piperacillin-tazobactam	4.5 g	Q8H	4 days after source control OR 8 days in Critical Illness	N/A	Levofloxacin and Metronidazole Ertapenem
Diverticulitis without Source Control	Piperacillin-tazobactam	4.5 g	Q8H	8 days	N/A	Levofloxacin and Metronidazole Ertapenem
Cholecystitis	Piperacillin-tazobactam	4.5 g	Q8H	Post Cholecystectomy with local inflammation: 24 hours Perforated with source control (includes PCT): 4 days Perforated without source control: 8 days	N/A	Levofloxacin and Metronidazole
Small bowel Obstruction with Necrosis or Bowel Resection	Piperacillin-tazobactam	4.5 g	Q8H	4 days after source control OR 8 days in Critical Illness	N/A	Levofloxacin and Metronidazole
Small bowel Obstruction without Necrosis or Bowel Resection	Cefazolin <i>plus</i> Metronidazole	2 g 500 mg	Q8H Q12H	24 hours	N/A	Levofloxacin and Metronidazole
Nectrotizing Soft Tissue Infection	Linezolid <i>plus</i> Piperacillin-tazobactam	600 mg 4.5 g	Q12H Q8H	48-72 hours after source control is achieved	N/A	The addition of clindamycin is recommended when there is strong suspicion for Group A Streptococcus or Toxic Shock Syndrome.
Perforated Peptic Ulcer Disease	Piperacillin-tazobactam +/- Antifungal Coverage in high risk/immunocompromised patients	4.5 g	Q8H	4 days after source control OR 8 days in Critical Illness	N/A	Levofloxacin and Metronidazole +/- Antifungal Coverage in high risk/immunocompromised patients: micafungin 100 mg Q24H
Elective Surgery						
Cholecystectomy	Cefazolin	2 g	One-time dose	1 dose	Every 4 hours	N/A
Inguinal Hernia	Cefazolin	2 g	One-time dose	1 dose	Every 4 hours	Vancomycin x 1 dose if known MRSA colonization
Ventral Hernia	Cefazolin	2 g	One-time dose	1 dose	Every 4 hours	Vancomycin x 1 dose if known MRSA colonization
Colectomy	Cefazolin <i>plus</i> Metronidazole	2 g 500 mg	One-time dose	1 dose	N/A	N/A
Colostomy/Ileostomy Reversal	Cefazolin <i>plus</i> Metronidazole	2 g 500 mg	One-time dose	1 dose	N/A	Vancomycin x 1 dose if known MRSA colonization <i>plus</i> Metronidazole
Cardiac & Thoracic	Cefazolin	2 g	One-time dose	1 dose	Every 4 hours	Vancomycin x 1 dose if known MRSA colonization
Skin & Soft Tissue	Cefazolin	2 g	One-time dose	1 dose	Every 4 hour	Vancomycin x 1 dose if known MRSA colonization
ICU & General Floor						
Hospital Acquired or Ventilator Associated Pneumonia (HAP/VAP)	Vancomycin <i>plus</i> Piperacillin-tazobactam	Pharmacy to dose 4.5 g	Pharmacy to dose Q8H	7days	N/A	Linezolid
Bacteremia	Vancomycin <i>plus</i> Piperacillin-tazobactam	Pharmacy to dose 4.5 g	Pharmacy to dose Q8H	Depends on Source and Isolated Bacteria	N/A	Linezolid Antifungal Coverage in high risk/immunocompromised patients: micafungin 100 mg Q24H
Complicated Urinary Tract Infection/CAUTI/Urologic Instr	Ceftriaxone	2 g	Q24H	7 days	N/A	N/A
Uncomplicated Urinary Tract Infection	Sulfamethoxazole-TMP	800 mg/160 mg	Q12H	3 days	N/A	N/A
Sinusitis	Ampicillin-sulbactam	3 g	Q6H	5 days	N/A	N/A
Cellulitis - Purulent (or MRSA risk factors)	Vancomycin	Pharmacy to dos	Pharmacy to dose	5-10 days	N/A	Linezolid
Cellulitis - Non-Purulent	Cefazolin	2 g	Q8H	5 days	N/A	N/A



Musculoskeletal Open extremity fractures

Timing and sequence for the treatment of femoral diaphyseal fractures in multiply injured patients:

Hemodynamically stable patients

- External fixation vs definitive internal fixation within 24 hours of clearance by the trauma team.

Hemodynamically unstable patients

- Temporary traction vs reduction/splinting for all fractures in the Emergency Department
- External fixation at first OR visit or once cleared by the trauma team

Open extremity fractures:

Classification:

Gustilo-Anderson Classification of Open Fractures	Description
Type I	Open fracture with a wound <1cm long, low energy, without gross contamination
Type II	Open fracture with a wound 1-10cm long, low energy, without gross contamination or extensive soft-tissue damage, flaps, or avulsions
Type III	IIIA: Open fracture with a wound greater than 10cm with adequate soft-tissue coverage, or any open fracture due to high-energy trauma or with gross contamination, regardless of the size of the wound IIIB: Open fracture with extensive soft-tissue injury or loss, with periosteal stripping and bone exposure that requires soft-tissue coverage in the form of muscle rotation or transfer. IIIC: Open fracture associated with arterial injury <i>requiring repair</i>.

Antibiotics:

- Timing: Administer within **30 min** of arrival to the ED & all patients to have MRSA nasal screen
- Antimicrobial:
 - Type I & II - **Cefazolin** 2g IV q8 hours
 - If penicillin allergy: **Vancomycin**** 15mg/kg IV q12 hours, Pharmacy To Dose Consult
 - Type III - Treatment for ALL Type III: : **Cefepime*** 2g q8 hours & **Vancomycin**** 15mg/kg IV q12 hours Pharmacy To Dose Consult



- Antimicrobial Cont.

- Soil, Fecal Contamination, and standing water contamination – **Cefepime*** 2g q8 hours, **Vancomycin**** 15mg/kg IV q12 hours Pharmacy To Dose Consult, & **Metronidazole** 500mg IV Q8h

Antimicrobial	Adult Dosing (age ≥15 years)	Pediatric Dosing
Cefazolin*	2 g IV Q8h	< 60 kg: 33 mg/kg/dose IV Q8h ≥ 60 kg: 2 g IV Q8h
Cefepime*	2 g IV Q8h	< 40 kg: 50 mg/kg/dose IV Q12h ≥ 40 kg: 2 g IV Q12h
Metronidazole	500 mg IV Q8h	< 37.5 kg: 13.3 mg/kg/dose IV Q8h ≥ 37.5 kg: 500 mg IV Q8h
Vancomycin**	15 mg/kg IV Q12h (max 1.5g/dose) Vancomycin IV - Pharmacy to Dose	20 mg/kg/dose IV Q8h (max 1.5g/dose) Vancomycin IV - Pharmacy to Dose

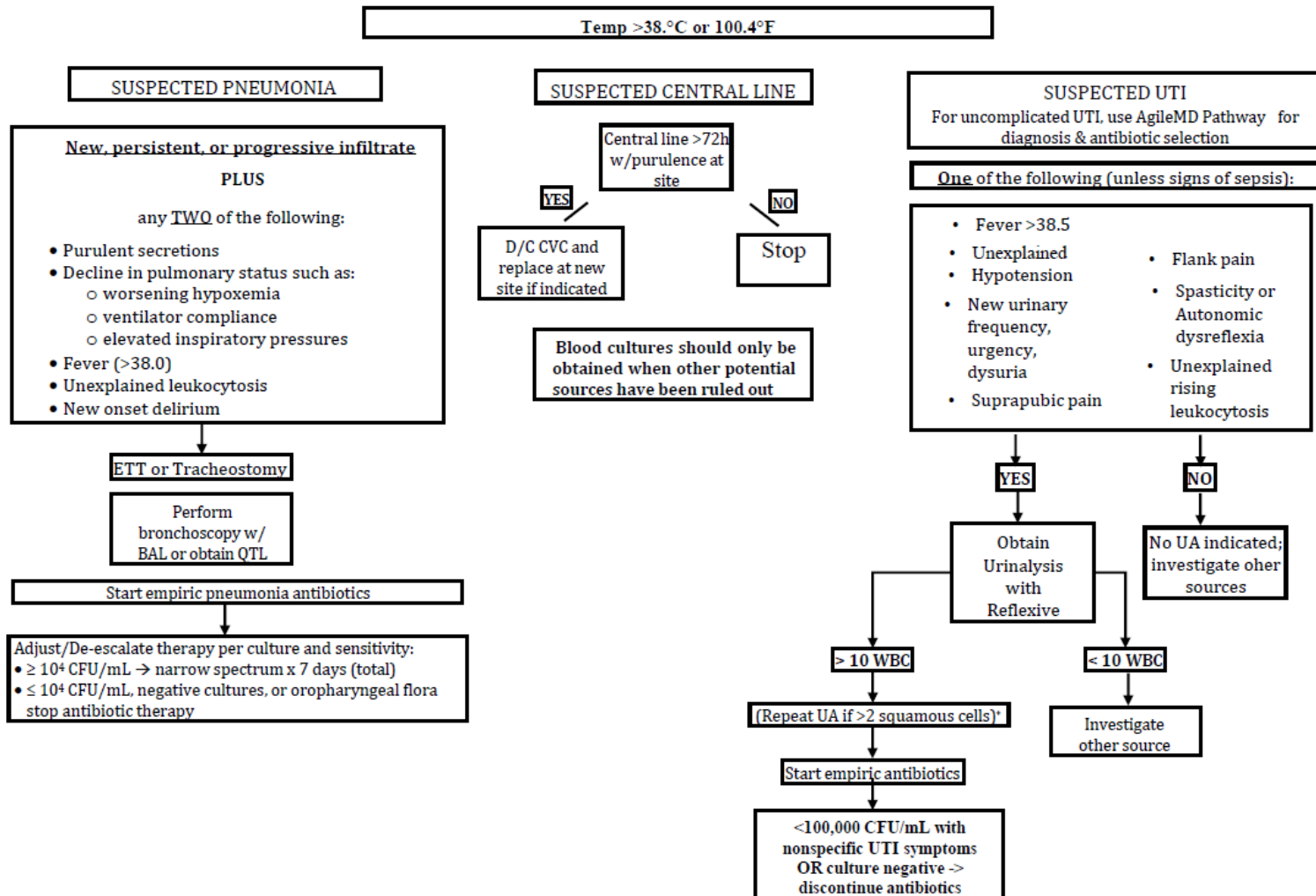
*Pharmacy to dose adjust per renal dosing protocol

**Order Vancomycin IV – Pharmacy to Dose for adult and pediatric patients. Suggested dosing listed above. Pharmacy to adjust based on renal function.

- Duration:
 - Type I & Type II Fractures – 24 hours
 - Type III – 24 hours after closure or 72 hours (whichever precedes)
- Operative debridement within 24 hours:
 - Gustilo and Anderson IIIC (vascular injury requiring repair) constitutes an emergency, should be at OR staging within 2 hours.
 - Type I to IIIB within 12 – 24hrs based on OR availability and patient stability.



Infectious Workup and Antimicrobial Guideline



Reference Standards:

[NHSN Hospital Acquired Pneumonia & VAP](#)

[NHSN Catheter Associated Urinary Tract Infection](#)

[NHSN Central Line Associated Blood Stream Infection](#)

GRAM POSITIVE BACTERIA

All ages. All body sites.

University of Missouri – University Hospital (Excluding Womens & Childrens and Clinics)

JANUARY 1-DECEMBER 31, 2024

	# of Isolates	Ampicillin	Oxacillin	Penicillin	Cefotaxime	Vancomycin	Levofloxacin	Clindamycin	Daptomycin	Linezolid	Nitrofurantoin (Urinary Isolates only)	Rifampin ^	Sulfamethoxazole- Trimethoprim	Erythromycin	Tetracycline
<i>Enterococcus faecalis</i>	292	100				98			100	100	100				
<i>Enterococcus faecium</i>	66	25				52			100	100					
<i>Staphylococcus aureus</i>	915														
<i>Staphylococcus aureus</i> (Methicillin Susceptible- MSSA)	496		100			100		93					97		93
<i>Staphylococcus aureus</i> (Methicillin Resistant- MRSA)	419					100		65	100	100		100	75		89
<i>Staphylococcus epidermidis</i>	366		30			100		48							78
<i>Staphylococcus lugdunensis</i>	34		88			100	97								94
<i>Staphylococcus capitis</i>	46		83			100		91							
<i>Streptococcus agalactiae</i>	73			100		100		32							
<i>Streptococcus anginosus</i>	62			98		100		65							
<i>Streptococcus pneumoniae (Non-CNS)</i>	78			96	100		100						76	52	
<i>Streptococcus pyogenes</i>	43			100		100		90						90	

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Blank spaces indicate antibiotics not tested. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalexin.

^Rifampin is 100% susceptible against all *Staphylococcus aureus* strains. Use only as adjunct to main therapy.

GRAM NEGATIVE BACTERIA

All ages. All body sites.

University of Missouri – University Hospital (Excluding Womens & Childrens and Clinics)

JANUARY 1-DECEMBER 31, 2024

	# of Isolates	Ampicillin	Ampicillin-Subactam	Piperacillin-Tazobactam	Cefazolin (Urinary Isolates only)	Cefoxitin	Ceftazidime	Ceftriaxone	Cefepime	Meropenem	Amikacin	Gentamicin	Tobramycin	Ciprofloxacin	Levofloxacin	Nitrofurantoin (Urinary Isolates only)	Sulfamethoxazole- Trimethoprim	Minocycline
<i>Citrobacter freundii</i>	48	•	•	•	•	•	•	•	98	94		88	94	83		83		
<i>Citrobacter koseri</i>	30			93				93	93			93		93				
<i>Enterobacter cloacae</i>	142	•	•	•	•	•	•	•	97	97		97	93	91	92		87	
<i>Escherichia coli</i>	1379	50	76	98	94	93	96	89	98	100	99	91	92	77	71	99	87	
<i>Klebsiella aerogenes</i>	50	•	•	•	•	•	•	•	100	90		100	100	100	95		100	
<i>Klebsiella oxytoca</i>	49		84					96	98	100		96		94	94		92	
<i>Klebsiella pneumoniae</i>	334		82	98	98	95	92	90	99	99	100	95	96	84	78	54	82	
<i>Morganella morganii</i>	33			100				100	100	100		91		78			76	
<i>Proteus mirabilis</i>	202	74	88	100	90	100	100	92	97	99	100	91	88	65	66		72	
<i>Pseudomonas aeruginosa</i>	342			97			94		95	93	99 [△]	▲		88	81			
<i>Raoultella ornithinolytica</i>	49		90					100	100	100				98			98	
<i>Serratia marcescens</i>	74							99	100	100		97		99	96		100	
<i>Stenotrophomonas maltophilia</i>	50														94		94	100

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Blank spaces indicate antibiotics not tested.

● HECK-Yes bacteria include *Hafnia alvei*, *Enterobacter spp.*, *Citrobacter freundii*, *Klebsiella aerogenes*, and *Yersenia enterocolitica*, because of the potential for inducible AmpC mediated beta-lactamase production, these organisms should be considered to be resistant to these antimicrobials. Bacteria producing inducible AmpC-resistance enzymes can degrade 1st, 2nd, & 3rd generation cephalosporins, penicillins, and overcome beta-lactamase inhibition by piperacillin-tazobactam. Although these bugs may show 'susceptible' upon initial testing, resistance can develop as early as a single day after start of drug therapy.

▲ Gentamicin used for *Pseudomonas* is no longer recommended by Clinical and Laboratory Standards Institute.

△ Amikacin used for *Pseudomonas aeruginosa* should only be used for urinary tract infections.

Emergency Department (Main)
All ages. All body sites.
University of Missouri – University Hospital
JANUARY 1-DECEMBER 31, 2024

		# of Isolates	Ampicillin	Ampicillin-Sulbactam	Cefazolin	Cefepime	Ceftazidime	Ceftioxone	Ciprofloxacin	Clindamycin	Levofloxacin	Meropenem	Nitrofurantoin (Urinary Isolates Only)	Oxacillin	Penicillin	Piperacillin-Tazobactam	Rifampin	Sulfamethoxazole- Trimethoprim	Tetracycline	Amikacin	Gentamicin	Tobramycin	Vancomycin
Gram Positive	<i>Enterococcus faecalis</i>	111	100																				100
	<i>Staphylococcus aureus</i>	311																					
	<i>Staphylococcus aureus</i> (Methicillin Susceptible- MSSA)	158			100					81				100				96	95				100
	<i>Staphylococcus aureus</i> (Methicillin Resistant- MRSA)	153								70							100	71	90				100
	<i>Staphylococcus epidermidis</i>	173								70				31					77				100
	<i>Streptococcus pyogenes</i>	31								87					100								100
Gram Negative	<i>Enterobacter cloacae</i>	54	•	•	•	94	•	•	85					•	•	•		80			94	94	
	<i>Escherichia coli</i>	938	50	76	95	99	96	89	75		62	100	96			96		74		99	90	91	
	<i>Klebsiella pneumoniae</i>	176		85	98	98	93	89	86			100				95		86		100	96	96	
	<i>Proteus mirabilis</i>	119	73	85	88	98	100	91	64			100				100		73			92	89	
	<i>Pseudomonas aeruginosa</i>	106				93	94		82		74	92				99				100 [△]	▲	100	

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Blank spaces indicate antibiotics not tested. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalexin.

^ Use only as adjunct to main therapy.

▲ Gentamicin used for *Pseudomonas* is no longer recommended by Clinical and Laboratory Standards Institute.

△ Amikacin used for *Pseudomonas aeruginosa* should only be used for urinary tract infections.

• HECK-Yes bacteria

Intensive Care Units

All ages. All body sites.

University of Missouri – University Hospital (CICU, MICU, SICU, NSICU, and PICU)

JANUARY 1-DECEMBER 31, 2024

		# of Isolates	Ampicillin	Ampicillin-Sulbactam	Cefazolin	Cefepime	Ceftazidime	Ceftriaxone	Ciprofloxacin	Clindamycin	Daptomycin	Erythromycin	Levofloxacin	Linezolid	Meropenem	Minocycline	Nitrofurantoin (Urinary Isolates Only)	Oxacillin	Penicillin	Piperacillin-Tazobactam	Rifampin ^	Sulfamethoxazole- Trimethoprim	Tetracycline	Amikacin	Gentamicin	Tobramycin	Vancomycin
Gram Positive	<i>Enterococcus faecalis</i>	56	100																								96
	<i>Staphylococcus aureus</i>	236																									
	<i>Staphylococcus aureus</i> (Methicillin Susceptible- MSSA)	148			100					94	100							100				99	90				100
	<i>Staphylococcus aureus</i> (Methicillin Resistant- MRSA)	88								58				100							100	79	90				100
	<i>Staphylococcus epidermidis</i>	93								44								29					81				100
	<i>Streptococcus pneumoniae</i>	37										49	100						100			76					
Gram Negative	<i>Enterobacter cloacae</i>	42	•	•	•	100	•	•	100				95					•	•	•		93			100		
	<i>Escherichia coli</i>	127	53	76	95	99	100	88	73				67		99		96			94		79			94	93	
	<i>Klebsiella pneumoniae</i>	61		80		98		91	88				85		100					95		72			97		
	<i>Pseudomonas aeruginosa</i>	107				94	94		82				76		94					95				97	▲		

General Notes:

Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Blank spaces indicate antibiotics not tested. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalixin.

^ Use only as adjunct to main therapy.

▲ Gentamicin used for *Pseudomonas* is no longer recommended by Clinical and Laboratory Standards Institute.

• HECK-Yes bacteria

Candida Species
All ages. All body sites.
University of Missouri – University Hospital (Excluding Clinics)
JANUARY 1 2021-DECEMBER 31, 2024

	# of Isolates	Caspofungin [§]	Fluconazole
<i>Candida albicans</i>	86	100	100
<i>Candida parapsilosis</i>	38	100	88
<i>Candida glabrata</i>	10*	100	
<i>Candida tropicalis</i>	20*	100	93
<i>Candida krusei</i>	6*	100	
<i>Candida dubliniensis</i>	5*	100	100
<i>Candida guilliermondii</i>	2*	100	
<i>Candida pelliculosa</i>	1*	100	100
General Notes: Percent (%) susceptible does not include duplicate isolates or intermediate susceptibility results. Blank spaces indicate antibiotics not tested.			
*Data for fewer than 30 isolates per organism, per selected time frame is not statistically significant according to CLSI standards.			
Yeast antifungal susceptibilities only performed per Micro policy (i.e. Sterile body fluids, physician requests, etc).			
§Caspofungin used to interpret susceptibility to Micafungin. Echinocandins are NOT to be used for CNS and urinary infections			

JANUARY 1, 2021-DECEMBER 31, 2024

Blank spaces indicate antibiotics not tested. Oxacillin susceptibility indicates susceptibility to cefazolin & cephalixin.



Reference

1. Jones, B. E., Ramirez, J. A., Oren, E., et al. (2025). *Diagnosis and management of community-acquired pneumonia: An official American Thoracic Society clinical practice guideline. American Journal of Respiratory and Critical Care Medicine*. <https://doi.org/10.1164/rccm.202507-1692ST>
2. Metlay, J. P., Eaterer, G. W., Long, A. C., et al. (2019). Diagnosis and treatment of adults with community-acquired pneumonia: An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America. *American Journal of Respiratory and Critical Care Medicine*, 200(7), e45–e67.
3. Fagon, J. Y., Chastre, J., Wolff, M., et al. (2000). Invasive and noninvasive strategies for management of suspected ventilator-associated pneumonia: A randomized trial. *Annals of Internal Medicine*, 132, 621–630.
4. Guidry, C. A., Mallicote, M. U., Petroze, R. T., Hranjec, T., Rosenberger, L. H., Davies, S. W., & Sawyer, R. G. (2014). Influence of bronchoscopy on the diagnosis of and outcomes from ventilator-associated pneumonia. *Surgical Infections*, 15(5), 527–532.
5. Kalil, A. C., Metersky, M. L., Klompas, M., et al. (2016). Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clinical Infectious Diseases*, 63(5), e61–e111.
6. Sharp, J. P., Magnotti, L. J., Weinberg, J. A., Swanson, J. M., Schoreppel, T. J., Clement, L. P., Wood, G. C., Fabian, T. C., & Croce, M. A. (2015). Adherence to an established diagnostic threshold for ventilator-associated pneumonia contributes to low false-negative rates in trauma patients. *The Journal of Trauma and Acute Care Surgery*, 78(3), 468–474.
7. Younan, D., Delozier, S. J., Adamski, J., Loudon, A., Violette, A., Ustin, J., Tinkoff, G., Moorman, M. L., McQuay, N., & UHRISES Research Consortium. (2020). Factors predictive of ventilator-associated pneumonia in critically ill trauma patients. *World Journal of Surgery*, 44(4), 1121–1125. <https://doi.org/10.1007/s00268-019-05286-3>
8. Joung, M. K., Lee, J., Moon, S. Y., et al. (2011). Impact of de-escalation therapy on clinical outcomes for intensive care unit-acquired pneumonia. *Critical Care*, 15(2), R79.
9. De Bus, L., Depuydt, P., Steen, J., et al. (2020). Antimicrobial de-escalation in the critically ill patient and assessment of clinical cure: The DIANA study. *Intensive Care Medicine*, 46(7), 1404–141
10. Bultas, A. C., Bery, A. I., Deal, E. N., Hartmann, A. P., Richter, S. K., & Call, W. B. (2019). Predictors of treatment failure following de-escalation to a fluoroquinolone in culture-negative nosocomial pneumonia. *Annals of Pharmacotherapy*, 53(12), 1207–1219.
11. Raman, K., Nailor, M. D., Nicolau, D. P., Aslanzadeh, J., Nadeau, M., & Kuti, J. L. (2013). Early antibiotic discontinuation in patients with clinically suspected ventilator-associated pneumonia and negative quantitative bronchoscopy cultures. *Critical Care Medicine*, 41(7), 1656–1663. <https://doi.org/10.1097/CCM.0b013e318287f713>
12. Vanderbilt University Medical Center, Division of Acute Care Surgery. (2025). *Infectious workup and antimicrobial stewardship guideline*. <https://www.vumc.org>
13. UTHHealth Houston. (2022, February). *Antibiotic therapy*. McGovern Medical School, Department of Surgery. <https://med.uth.edu/surgery/antibiotic-therapy/>